

Course: COS 506**Course Title: Advanced Programming/Data Structures** (3 credit hours)**INSTRUCTOR**

Professor	Xinxing Wu
Office	Hathaway Hall 307
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COURSE DETAILS

Semester	Spring 2026
Class Time	9:30AM – 10:45 AM (MW)
Classroom	Hathaway Hall 320
Office Hours	Posted on Office Door or by Appointment

University Catalogue Description

This course teaches students the advanced skills in programming and data structures.

Prerequisite: COS 502 or consent of instructor.

Course Objectives

This course reviews and introduces advanced object-oriented programming concepts, such as inheritance, abstract base classes, polymorphism, and dynamic binding. It also introduces O-O design techniques such as Abstract Data Types and Design by Contract. Those techniques will then be used to discuss, implement, and use various data structures like stacks, queues, lists, heaps, trees, and hash maps.

Learning Outcomes

THE FOLLOWING LIST ARE THE MASTER IN COMPUTER SCIENCE PROGRAM OUTCOMES:

1. Ability to apply reasoning, problem solving, technical skills, and discipline tools to solve a problem or issue with minimal guidance.
2. Ability to communicate theories, concepts and results to a technical audience in the form of a technical report and/or an oral presentation to a review committee.
3. Ability to participate and collaborate in a team environment to accomplish a common goal.
4. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Competency Levels

THE FOLLOWING LIST ARE THE STUDENT LEARNING OUTCOMES FOR THIS COURSE AND HOW THEY ARE MAPPED TO THE OVERALL PROGRAM OUTCOMES MENTIONED ABOVE, ALONG WITH THEIR LEVEL OF COMPETENCY. THE COMPETENCY LEVELS ARE:

I Introduced **R** Reinforced **M** Mastered

UPON COMPLETION OF THIS COURSE, STUDENTS WILL BE ABLE TO:

- Apply computer science theory and software development fundamentals to produce computing-based solutions. **M** (1,4)
- Students will implement, test, analyze, and measure the runtime behavior of particular data structures. **M** (1,4)
- Students will design and implement an intermediate programming project, that utilizes abstract base classes and data structures. **M** (1,4)

Course Thoro10

1. Communicate Effectively.
2. ☒ Think critically to solve problems and create new ideas and solutions.
3. Apply quantitative reasoning skills to analyze and solve numerical problems.
4. Interact effectively with people from different backgrounds.
5. Adapt to changing circumstances while leading and supporting others.
6. Perform professionally.
7. Engage in civic life to improve society.
8. Collaborate and work in teams.
9. Apply academic knowledge, skills, and abilities.
10. Use information for decision making.

Hallmark Assignment - **Final Project**: Practical "real-world" application Demo and Presentation

- CPE Essential Skills #2 will be assessed using the **AACU VALUE Rubric: Problem Solving (Thoro 10)** Rubric.
- Thoro10 Essential Skills: 2. Think critically to solve problems and create new ideas and solutions.
 - Through the final project demo and presentation, students will demonstrate critical thinking by evaluating assumptions, analyzing information, and drawing well-reasoned conclusions. They will also demonstrate creative thinking by integrating ideas in original ways and developing innovative solutions to real-world problems.
- **Hallmark assignment** in Canvas: [Link](#)

Required Texts

Other resource: Data Structure and Algorithms Visualizations

Course Calendar/Schedule

The course schedule is tentative and may be adjusted based on practical class needs. 😊 The instructor reserves the right to make changes or impose additional requirements as necessary.

Note: Artificial Intelligence (AI) Tool Usage - If you use any AI tool at any point in your work, you must clearly and fully disclose its use in your assignment, report, project submission, or any other deliverable. Your disclosure must be specific and detailed, including 🗣️

- Where AI was used (e.g., summarizing readings, generating ideas, code assistance, debugging, data analysis, data visualization, outlining, editing).
- What tool(s) were used (tool name and, when available, version/model, plus platform/app).
- Why you chose to use the tool (e.g., brainstorming, translation, grammar checking, accelerating coding, improving clarity).
- How you used the output (e.g., modified/edited, cited/quoted, directly included, used only as a reference, used to compare against your own work).
- Other details as needed. 😊 The instructor reserves the right to impose additional disclosure requirements as necessary.

🔔 Important: Failure to disclose AI use—or presenting AI-generated work as entirely your own without clear clarification—may be considered a violation of academic integrity.

GRADING SCALE

This grading scale outlines how final grades are determined based on the total points earned in the course. Each letter grade corresponds to a specific range of percentages or points.

Course Specific Requirements, Expectations, Policies

Class Attendance

Students are expected to attend the entire period of each class throughout the semester. The university attendance policy is described in the 2022-2023 Kentucky State University Handbook (<https://tinyurl.com/2022-2023student>). Students with service-related absences (military, on or off campus representation of the University, jury-duty) and excused absences (documented illness, documented illness or death of immediate family, court appearances, or university sponsored events/trips) should file documentation with the Dean of Student Leadership, Conduct, and Health in the Office of Student Engagement and Campus Life, housed in the Hill Student Center, Office 334.

Unexcused absences: To help maintain a smooth learning environment, students who arrive more than five (5) minutes late to start of class AND OR leaving class early will be marked as 'Absent'. Each three (3) unexcused absences will be a drop in a letter grade for the course. If you have any form of note or excuse IT must be submitted to Student Services for review and dispersion! No exceptions!

Cell Phone Policy

To create a focused and respectful learning environment, all cell phones, PDAs, beepers, and similar electronic devices must be turned off and put away during class. Texting is not permitted at any time during class. Failure to follow this policy will result in an **unexcused absence** for that day.

Headphones, earbuds, and music-playing devices may not be worn or used during class. During exams, no cell phones may be on desks or visible at any time; the presence of a cell phone during a test will result in a zero for that exam.

Assignments

Classwork and homework may be assigned and may serve as the basis for in-class discussions. To support meaningful learning, students are expected to come to class prepared and engaged, with the goal of developing a thorough understanding of the course material. A due date will be provided on Canvas or announced in class for each weekly homework assignment. All assignments are due by the end of the day on the posted due date. Please pay close attention to assignment and exam deadlines. In general, deadlines will not be extended, and late submissions are not accepted.

Generally, assignments are required to be submitted to Canvas. Do not use KSU email to email/submit your work.

Excused Absences

For any excused absence, all make-up assignments must be completed within one week.

Student Responsibilities

If you need to miss a class, please remember that you are responsible for keeping up with course expectations. This includes finding out what assignments were given, noting any announced due dates, and reviewing the material covered during your absence.

Use of Instructional Technology

Kentucky State University supports instructional uses of technology in the classroom for classroom-based purposes at the instructor's discretion and direction. (See the student handbook at <https://tinyurl.com/2022-2023student>)

Canvas

Course Syllabus will be posted on the course Canvas site (<https://kysu.instructure.com/>).

The materials provided on Canvas are intended for educational use only. Please do not share, upload, or distribute these materials on the internet or to others outside this course.

For other tutorials over other KSU software systems (e.g. SafeAssign, Upswing, etc.), visit the Kentucky State University Distance Education Online Resource (<https://tinyurl.com/KSUonlineresources>).

Additional Resources

- **Paul G Blazer Library:** KSU Library main page (<https://www.kysu.edu/academics/library/>)
- **The Thoroughbred Learning Center (TLC):** The Thoroughbred Learning Center (<https://tinyurl.com/KSUTLC>) is an academic support unit designed to facilitate student success, retention, and graduation. The Center provides comprehensive learning assistance to meet the academic demands of all students along all points of their academic careers. The TLC Center is located on the 4th floor of the Julian M. Carroll ASB building, Suite 406 and can be reached by emailing tlc@kysu.edu or calling 502-597-5599.

- Students can stay current on activities and services provided by the TLC Center by visiting the TLC Instagram account (https://www.instagram.com/kysu_tlc/), as well.

Academic Honesty

All students must conduct their studies at KSU honestly, ethically and in accordance with accepted standards of academic conduct. See the Academic Conduct section of the 2022-2023 KSU Student Handbook (tinyurl.com/2022-2023student) (pp. 19-21). Cheating and plagiarism will result in a score of zero on that particular test or assignment and thus lowering of grade in the course.

- **Cheating:** Cheating is defined as the fraudulent or deceptive taking, giving, or presenting of any information or material with the intent of aiding oneself or another on any academic work which is considered in any way in the determination of a course grade. Cheating includes but is not limited to copying a hand-in assignment, looking on another person's exam during a test, signing in for someone else, inappropriate use of technology during an exam (like having a cell phone on during a test), and the un-permitted use of "cheat sheets" or using internet sites.

Copying and pasting from the internet is considered cheating as you are using the intellectual property of someone else and turning it in as you own work. In addition, **the use of AI to complete assignments and or submissions will not be tolerated, except for special circumstances explicitly stated earlier, any use of AI tools must be fully disclosed.**

Cheating may result in an F in the class. It may also result in expulsion from the university.

In addition,

- Documentation of cheating will be provided by the instructor to the school chair, college dean, and registrar for implementing into student record. It will be up to the instructor if a second attempt is made possible for completing that exam, assignment or activity.
- Bathroom breaks will not be allowed during exams/tests. Any student leaving the classroom or lab during an exam/test. The exam/test will be turned in with no additional time or retake.
- No cell phones, ear buds, earmuffs, large coats, head coverings, caps or hats will be allowed during exams/tests.
- Not following these stipulations, you may be asked to leave the classroom or lab.
- **Plagiarism:** Plagiarism is the intentional or unintentional act of submitting the work of another as one's own. It includes the submission of the complete or partial work, or of the words, ideas, or format of another, published or unpublished, without appropriate reference and source credit. It includes employing or permitting another person to produce, alter, or revise material which the student submits as his or her own. If the student has any uncertainty regarding plagiarism in the submission of any material to his or her instructor, he or she should discuss the matter with the instructor prior to submission of the material.
- **Forgery:** Forgery is falsely recording the signature of an advisor, instructor, or any university official on any official University academic document.

Notice to Students with Disabilities

The mission of the Testing and Disability Services is to facilitate equal access to KSU's campus, programs, activities, and services, in accordance with federal and state regulations and University policies. We support this mission by applying eligibility guidelines and coordinating the delivery of appropriate accommodations so that students with disabilities can pursue their educational goals. Additionally, we promote greater disability awareness campus wide. The University also provides reasonable accommodations to students and campus visitors with disabilities through Testing and

Disability Services in Hathaway Hall room 107. Students with disabilities are invited to contact Dr. Phillip Clay in the DRC at 502-597-6041 for information on eligibility, required documentation, and available services. Additional information concerning the DRC and accommodations can be found by visiting the Kentucky State University Office of Accessibility and Disability Services (<https://www.kysu.edu/academics/disability-svcs/index.php>).

Statement on Veterans Affairs

Kentucky State University's Veterans Affairs Coordinator provides information regarding educational benefits from the Department of Veterans Affairs (DVA), assists in applying for GI Bill© benefits, certifies enrollment to the Veterans Affairs (VA) for payment each semester, and assists in resolving payment problems. For more information about educational benefits programs for Veterans, contact the Department of Veterans Affairs (<https://benefits.va.gov/gibill/>) at 1-888-442-4551 and visit Kentucky State University's Veteran and Military Student Affairs (<https://www.kysu.edu/academics/vet-and-mil-affairs/index.php>) online.

Mission Statement

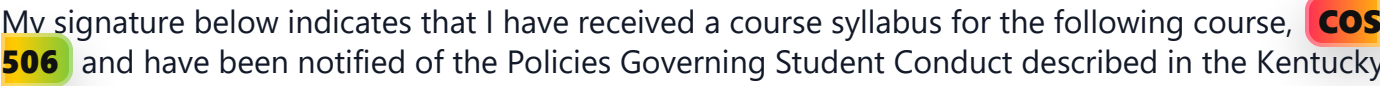
The Objectives and Learning Outcomes of this course directly support the Mission of Kentucky State University (<https://www.kysu.edu/about-ksu/mission-vision-core-values.php>).

Kentucky State University is a public, comprehensive, historically black land-grant university committed to advancing the Commonwealth of Kentucky, enhancing society, and impacting individuals by providing quality teaching with a foundation in liberal studies, scholarly research, and public service to enable productive lives within the diverse global economy.

Based on practical class needs, 😊 the instructor reserves the right to make changes or impose additional requirements as necessary.

  Syllabus Acknowledgement Form is to be completed and submitted the FIRST week of this course. NO Exceptions. Any student NOT submitting this form the FIRST week is applicable to be dropped from this course.

Student Acknowledgement of Receipt of Course Information

My signature below indicates that I have received a course syllabus for the following course,  and have been notified of the Policies Governing Student Conduct described in the Kentucky State University Student Handbook (<https://tinyurl.com/2022-2023student>) and Miscellaneous Policies in the University Catalogue (<https://www.kysu.edu/academics/registrar/select-a-catalog.php>).

I agree to read these documents, and I agree to sign and deliver this copy of the "Student Acknowledgment" form within two (2) weeks of the start of the semester. I understand that the policies contained within these documents apply directly to me and to all students in the class. I agree to abide by these policies and recognize that not abiding by these policies could result in dismissal from this class and/or affect my standing as a student at KSU as described in the Policies Governing Student Conduct in Student Handbook and Miscellaneous Policies in the University Catalogue.

Name (please print):

CWID / Student ID:

Date:

mm / dd / yyyy



CONTACT INFORMATION (PLEASE PRINT CLEARLY):

Local Address:

Local Phone:

E-Mail:

HANDWRITTEN SIGNATURE

Sign in the box below

 Signature (printed)

Please do not hesitate to reach out xinxing.wu@kysu.edu if you have any questions.